

Quantum-Assisted Multi-Asset Portfolio Construction

Step 04

Classical Constraint Ladder

Showing why unconstrained optima fail and how investable portfolios emerge

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1 Why a constraint ladder is useful

A single final optimization hides the reason a portfolio becomes investable. Step 4 solves six increasingly realistic problems. Each stage adds one family of constraints or costs, so the reader can see exactly which textbook solutions fail the final policy.

2 Analytical reference portfolios

With only full investment and a positive-definite covariance matrix, the global minimum-variance portfolio solves

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1. \quad (1)$$

The Lagrangian is

$$\mathcal{L} = w^\top \Sigma w + \nu(\mathbf{1}^\top w - 1). \quad (2)$$

Stationarity gives $2\Sigma w + \nu\mathbf{1} = 0$, hence

$$w_{\text{GMV}} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}. \quad (3)$$

This formula explains why unconstrained minimum variance can become extremely concentrated: it follows covariance geometry without any economic cap or implementation rule.

A mean-variance reference solves

$$\min_w \lambda w^\top \Sigma w - \mu^\top w \quad \text{s.t.} \quad \mathbf{1}^\top w = 1. \quad (4)$$

Its first-order condition is

$$2\lambda \Sigma w - \mu + \nu\mathbf{1} = 0. \quad (5)$$

Bounds and real portfolio rules destroy the simple closed form, motivating numerical quadratic programming.

3 Lifted variables and linear constraints

The numerical variable vector contains weights, purchases, sales, and optionally scenario excesses:

$$x = (w, p, n, h). \quad (6)$$

The accounting equality for each asset is

$$w_i - p_i + n_i = w_i^0. \quad (7)$$

All class, factor, return, income, turnover, capacity, and scenario constraints become rows of one linear system

$$\ell \leq Ax \leq u. \quad (8)$$

This is useful both computationally and for later independent reconstruction.

4 Stage objective and gradient

The objective used in the solver is

$$F(x) = \lambda_v w^\top \Sigma w - \alpha_g g^\top w - \alpha_y y^\top w + \lambda_c w^\top w \quad (9)$$

$$+ \lambda_L c^\top (p + n) + \lambda_I (w - w^0)^\top \Lambda (w - w^0) + \lambda_s \sum_s \omega_s h_s^2. \quad (10)$$

The analytical gradient is

$$\nabla_w F = 2\lambda_v \Sigma w - \alpha_g g - \alpha_y y + 2\lambda_c w + 2\lambda_I \Lambda (w - w^0), \quad (11)$$

$$\nabla_p F = \lambda_L c, \quad \nabla_n F = \lambda_L c, \quad (12)$$

$$\nabla_h F = 2\lambda_s (\omega \odot h). \quad (13)$$

Providing the gradient improves numerical reliability and makes the implementation auditable.

5 The six stages

1. **Minimum variance:** variance is minimized with basic bounds.
2. **Mean variance:** expected growth and income are rewarded.
3. **Asset caps:** concentration limits are activated.
4. **Guardrails:** class/factor bands and return/income floors are added.
5. **Trading costs:** exact turnover, liquidity capacity, linear cost, and nonlinear impact are added.
6. **Scenario aware:** warning slacks and hard scenario limits are added.

A stage may solve its own problem and still breach the final policy. The final-policy audit is therefore reported separately from solver success.

Stage	Return	Volatility	Gross turnover	Effective holdings	Max weight	Scenario penalty	Worst stress
01_minimum_variance	4.08%	0.45%	190.59%	2.09	50.88%	0.000000	0.05%
02_mean_variance	5.73%	3.81%	178.02%	2.16	64.95%	0.000000	6.46%
03_asset_caps	5.82%	5.93%	133.41%	10.99	15.00%	0.000000	10.63%
04_guardrails	5.80%	5.97%	110.60%	13.10	15.00%	0.007730	10.88%
05_trading_costs	5.44%	6.00%	50.00%	20.19	13.43%	0.002985	10.48%
06_scenario_aware	5.38%	5.94%	50.00%	20.70	13.43%	0.000025	10.36%

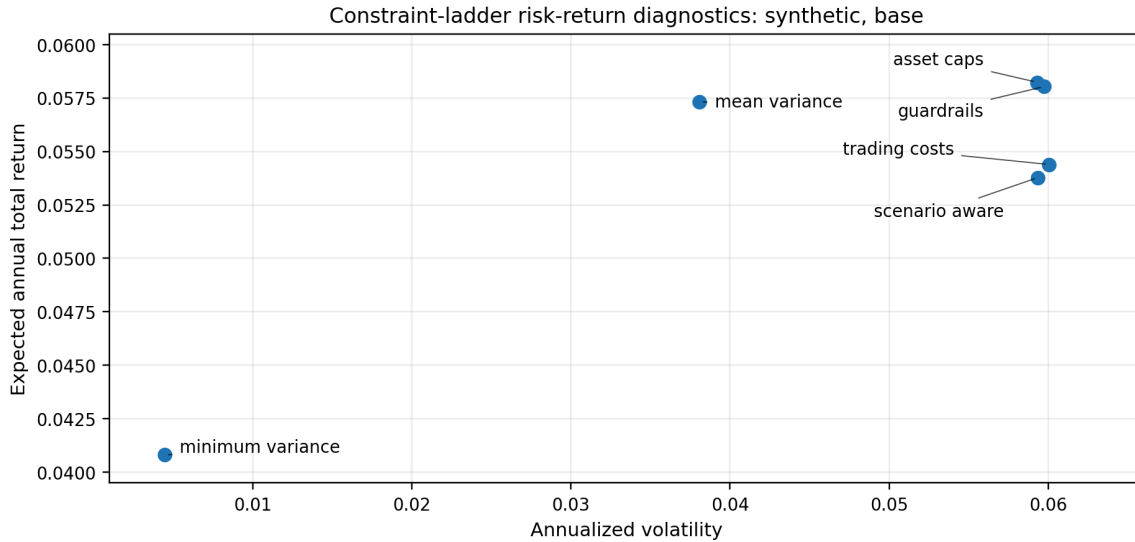


Figure 1: The constraint ladder exposes the cost of moving from mathematical optimality to investability.

6 Financial interpretation of the ladder

The minimum-variance portfolio has only 0.45% modeled volatility, but it uses 190.59% gross turnover, holds roughly two effective positions, and violates 13 hard rules. This is not a superior investment; it is a diagnostic showing that unconstrained covariance minimization exploits the model aggressively.

Asset caps increase effective holdings to about 11 and remove some breaches, but turnover remains 133.41%. Guardrails improve class balance, but turnover is still 110.60%. The trading-cost stage is the first to reach the 50% turnover limit and zero hard breaches. The scenario-aware stage then reduces the weighted scenario penalty from 0.002985 to 0.000025 while preserving hard compliance.

7 Scenario-aware result

Scenario	Loss	Warning	Hard limit	Warning excess	Hard pass
Global equity selloff	7.478%	11.410%	15.410%	0.000%	True
Inflation and rate shock	6.287%	6.972%	10.972%	0.000%	True
Credit and liquidity crisis	8.050%	9.909%	13.909%	0.000%	True
Commodity supply shock	2.144%	2.028%	6.028%	0.115%	True
Broad deleveraging shock	10.363%	12.327%	16.327%	0.000%	True

Only the commodity shock remains slightly above its warning threshold, by about 0.115 percentage point, while all five hard limits pass. This is the precise interpretation: hard-compliant with one soft warning breach.

8 Why the scenario-aware portfolio becomes important later

The scenario-aware Step 4 portfolio has approximately 5.38% expected return, 5.94% volatility, 10.36% worst stress loss, and 5.24% modeled drawdown in the later comparison data. These ratios are strong enough that Step 10 identifies it as the provisional zero-benchmark challenge-wide leader. However, it uses the full 50% turnover budget and was not included in the original Step 7 independent shortlist validation, so its universal certification remains pending.

9 Numerical solution and audit discipline

The original optimizer is SciPy SLSQP. Solver success is necessary but not sufficient. After every solve, the code recomputes purchases, sales, turnover, class exposures, factor exposures, expected return, income, capacity, and scenario limits. A stage is labeled successful only if both the numerical solver and the exact audit pass.

10 Reproduction checklist

Reproduce the six stages in order, warm-starting later stages from earlier solutions. Preserve the same objective coefficients and constraints; recompute exact warning excess rather than trusting auxiliary values; save every audit row; and distinguish stage-specific feasibility from final-policy feasibility.

11 What Step 4 proves

Step 4 proves that a classical convex workflow can construct zero-hard-breach portfolios and that realistic constraints materially change the solution. It does not prove that the scenario-aware stage is the final best portfolio under every mandate; Step 5 changes preferences, Step 6 compares candidates, Step 7 validates selected solutions independently, and Step 10 applies the challenge rule.

A Why a constraint ladder is more informative than one solve

A single final optimizer hides which financial rule caused the largest change. The ladder adds realism in stages and re-solves after each addition. This produces a causal narrative: minimum variance reveals concentration, mean-variance reveals turnover, asset caps reveal remaining allocation imbalances, guardrails expose implementation problems, trading constraints create an investable portfolio, and scenarios improve stress behavior.

Each stage is optimal only for its own declared problem. A successful solver status does not imply compliance with the later, stricter final policy. The project therefore applies the final-policy audit to every ladder portfolio and reports the breach count separately from solver success.

B Stage-by-stage mathematical specification

Let the core objective be

$$F(w, p, n, h) = -\alpha\mu^\top w + \lambda_v w^\top \Sigma w + \lambda_c w^\top w + \lambda_L c^\top (p + n) + \lambda_I \Delta w^\top \Lambda \Delta w + \lambda_s \sum_s \omega_s h_s^2. \quad (14)$$

The stages can be written as nested feasible sets $\mathcal{F}_1 \supseteq \dots \supseteq \mathcal{F}_6$ with changing objective terms:

1. **Minimum variance:** minimize $w^\top \Sigma w$ under budget and long-only constraints.
2. **Mean–variance:** add expected-return reward.
3. **Asset caps:** add $w_i \leq u_i$.
4. **Guardrails:** add class bands, factor bands, return and income floors.
5. **Trading costs:** add exact trade accounting, turnover, capacity, linear cost, and impact.
6. **Scenario aware:** add warning slacks and hard scenario limits.

The exact code uses a common solver function with switches controlling which terms and constraints are active. This avoids six unrelated implementations.

C Why minimum variance became concentrated

When correlations are strong, one low-volatility or defensive asset can dominate the variance solution. The minimum-variance stage reached approximately 0.45% modeled volatility but had only about 2.09 effective holdings and a maximum asset weight above 50%. Its gross turnover was approximately 190.59%. This is a mathematically valid variance optimum but not a valid implementation under the final mandate.

The lesson is general: variance alone rewards any concentration that happens to reduce the covariance quadratic form. Diversification and turnover must be stated explicitly if they matter.

D Why turnover can exceed 100 percent

Gross turnover is

$$T = \sum_i |w_i - w_i^0|. \quad (15)$$

Selling the entire current portfolio and buying a completely disjoint portfolio gives $T = 2$, or 200% gross turnover. Therefore the early-stage values above 100% are possible and correctly interpreted. One-way turnover is $T/2$.

E Scenario penalty derivation

For losses $L_s(w)$ and warning thresholds τ_s , define $h_s = \max(L_s(w) - \tau_s, 0)$. The weighted penalty is

$$P_{\text{scen}} = \sum_s \omega_s h_s^2. \quad (16)$$

The trading-cost stage had a scenario penalty around 2.985×10^{-3} , while the scenario-aware stage reduced it to roughly 2.5×10^{-5} . The expected return declined from about 5.439% to 5.378%, while volatility

and worst stress also improved. This is the intended trade-off: a small expected-return concession for substantially lower warning excess.

F Reading the ladder numerically

Stage	Return	Volatility	Gross turnover	Effective holdings	Final-policy breaches
Minimum variance	4.0806%	0.4468%	190.5944%	2.09	13
Mean–variance	5.7308%	3.8076%	178.0234%	2.16	8
Asset caps	5.8232%	5.9299%	133.4085%	10.99	4
Guardrails	5.8032%	5.9735%	110.5986%	13.10	1
Trading costs	5.4390%	6.0034%	50.0000%	20.19	0
Scenario aware	5.3780%	5.9354%	50.0000%	20.70	0

A reader should not treat this table as a monotonic performance contest. Each row answers a different question. The final two are the first rows that satisfy the full hard policy.

G Constraint audit construction

For every candidate and every hard rule, the audit stores

$$(\text{name, value, lower, upper, lower slack, upper slack, satisfied}). \quad (17)$$

A lower-bound rule passes when $v \geq l - \epsilon$ and an upper-bound rule passes when $v \leq u + \epsilon$. The tolerance is numerical, not economic; it should be small compared with reporting precision. The audit is recomputed from the final weights, purchases, sales, and scenarios rather than inferred from solver status.

H Warm starts and solver behavior

Each richer stage may use the previous stage as a warm start. A warm start can reduce solve time, but it must not be confused with a constraint. The new stage can move away from the prior solution. Because the continuous problem is convex under the declared construction, solver convergence plus independent feasibility checks is meaningful; Step 7 later performs a separate solver reconstruction.

I Bottlenecks encountered and why they matter

The project encountered practical issues that are typical in real optimization work:

- a stage could solve successfully while failing the final policy;
- scenario auxiliary variables could disagree with exact hinge values if not recomputed;
- transaction-cost parameters needed a higher-participation sensitivity case;
- warm-start and configuration objects had to remain consistent across notebook cells; and
- reporting needed to distinguish percentage values, decimal fractions, and basis points.

The final design handles these with a common metric function, a common audit table, exact scenario recalculation, and explicit configuration exports.

J **Reproduction protocol**

1. Load the validated Step 3 data and current portfolio.
2. Solve all six stages in order with the declared switches.
3. Recompute metrics from returned weights.
4. Apply the final-policy audit to every stage, not only the last stage.
5. Export weights, trades, metrics, and constraint rows.
6. Confirm the first four stages have the reported diagnostic breaches and the final two have zero hard breaches.
7. Confirm the scenario-aware stage reduces warning penalty without violating turnover, class, factor, return, income, liquidity, or hard scenario limits.

Final interpretation clarifications

These notes align interpretation with the final executed code path and saved evidence without changing the reported numerical results.

- Hard-breach counts reported for early ladder stages are final-policy audit counts, not violations of the smaller constraint set solved at that intermediate stage.
- Several research policy bands are centered around the synthetic reference incumbent to guarantee a controlled feasible demonstration. They are configurable research assumptions, not claimed Vanguard investment policy.
- The trading-cost and scenario-aware portfolios use the full 50% turnover allowance. Their later Step 10 strength should therefore be read together with the need for a tighter-turnover sensitivity study.
- The 5-day capacity window in Step 4 is a tradability constraint; Step 3's 3-day execution horizon is used in cost/impact calibration. The parameters serve different roles.